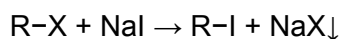


A group of students studied the reactivity of four alkyl halides (R-X) in two nucleophilic substitution reactions. Nucleophilic substitution reactions can go through one of two mechanisms. The S_N2 reaction requires substitution of a leaving group with a nucleophile in a single step (Reaction 1), and the S_N1 reaction requires elimination of the leaving group and formation of an intermediate before nucleophilic substitution can occur, resulting in a two-step process (Reaction 2).



Reaction 1

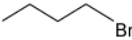
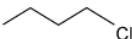
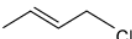


Reaction 2

Experiment 1

The students performed the S_N2 reaction (Reaction 1) by the addition of a few drops of each alkyl halide into a separate test tube containing 1 mL of 15% solution of NaI in acetone at 25°C. After mixing, they monitored the reaction for the formation of a precipitate (sodium halide salt). If no precipitate formed after 5 minutes, they raised the temperature of the reaction to 50°C and again checked for the formation of a precipitate. Formation of a precipitate indicated the substrate underwent nucleophilic substitution. The results observed for the Finkelstein reaction are shown in Table 1.

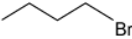
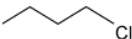
Table 1 Results of the S_N2 Reaction

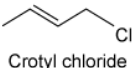
Substrate	Result at 25°C	Result at 50°C
 1-Bromobutane	Precipitate	—
 1-Chlorobutane	No reaction	Moderate precipitate
 <i>tert</i> -Butyl chloride	No reaction	No reaction
 Crotyl chloride	Precipitate	—

Experiment 2

The students repeated the procedure followed in Experiment 1 except they added a few drops of each alkyl halide into a separate test tube containing 1 mL of 1% solution of AgNO₃ in ethanol at 25°C. The results observed for this reaction are shown in Table 2.

Table 2 Results of the S_N1 Reaction

Substrate	Result at 25°C	Result at 50°C
 1-Bromobutane	No reaction	Trace precipitate
 1-Chlorobutane	No reaction	No reaction
 <i>tert</i> -Butyl chloride	Heavy precipitate	—

 Crotyl chloride	Heavy precipitate	—
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The substrate *tert*-butylchloride reacts in Experiment 2 but not in Experiment 1 because:

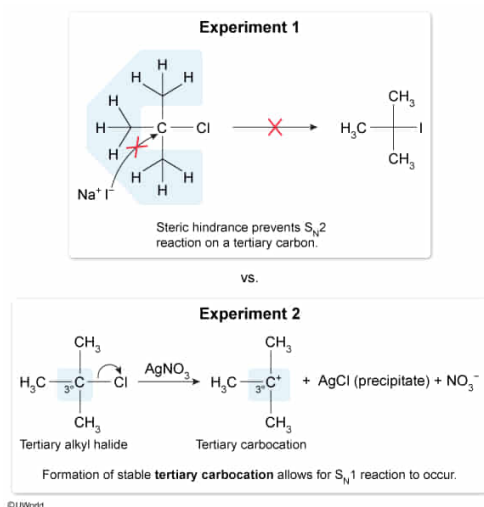
- I. the leaving group of *tert*-butylchloride is bonded to a primary carbon.
- II. a stable carbocation forms in Experiment 2.
- III. the leaving group of *tert*-butylchloride is bonded to a tertiary carbon.
- IV. an unstable carbocation forms in Experiment 2.

- ☐ A. I and II only [3%]
☐ B. I and IV only [2%]
☒ C. II and III only [85%]
☐ D. III and IV only [8%]

Correct

85% answered correctly

Explanation:



An S_N2 reaction is a nucleophilic substitution reaction that occurs in a single step whereas an S_N1 reaction occurs in **two steps**. In the first step of an S_N1 reaction, the bond between the leaving group and

carbon is broken, transferring the bonding electrons to the leaving group and forming a **carbocation** intermediate. This is the **rate-determining** (slowest) step. The second step involves formation of a bond between the nucleophile and the carbocation to form the final product of the reaction.

Tertiary alkyl halides undergo S_N1 reactions more readily than less substituted alkyl halides because **tertiary alkyl halides** form a more **stable carbocation**. Conversely, tertiary alkyl halides are unreactive in an S_N2 reaction because they are more **sterically hindered**, preventing nucleophilic attack and the breaking of the leaving group–carbon bond in one step.

The substrate *tert*-butyl chloride is a tertiary alkyl halide, meaning the chlorine leaving group is bonded to a carbon with three alkyl substituents (a tertiary carbon) (**Number III**). Therefore, *tert*-butyl chloride is sterically hindered and cannot undergo an S_N2 reaction with NaI, as described for Experiment 1.

Alternatively, breaking the bond between the chlorine and the tertiary carbon (the first step of an S_N1 reaction) causes a stable, tertiary carbocation to form (**Number II**), permitting *tert*-butyl chloride to go through an S_N1 reaction with $AgNO_3$ in Experiment 2.

(Number I) The leaving group of *tert*-butyl chloride is bonded to a tertiary, rather than a primary, carbon.

(Number IV) A carbocation *does* form in the first step of Experiment 2, but it is *stable* because the positive charge is on a tertiary carbon. S_N1 reactions are favored when stable carbocations can form.

Educational objective:

S_N1 reactions occur in two steps: carbocation formation and nucleophilic addition. These reactions readily occur with tertiary alkyl halides because they are sterically hindered and form stable carbocations.

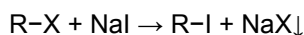
Time spent:0 sec

Subject:
Organic Chemistry

QID:401606

System:
Introduction to Organic Chemistry

A group of students studied the reactivity of four alkyl halides (R-X) in two nucleophilic substitution reactions. Nucleophilic substitution reactions can go through one of two mechanisms. The S_N2 reaction requires substitution of a leaving group with a nucleophile in a single step (Reaction 1), and the S_N1 reaction requires elimination of the leaving group and formation of an intermediate before nucleophilic substitution can occur, resulting in a two-step process (Reaction 2).



Reaction 1

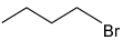
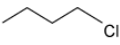
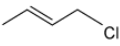


Reaction 2

Experiment 1

The students performed the S_N2 reaction (Reaction 1) by the addition of a few drops of each alkyl halide into a separate test tube containing 1 mL of 15% solution of NaI in acetone at 25°C. After mixing, they monitored the reaction for the formation of a precipitate (sodium halide salt). If no precipitate formed after 5 minutes, they raised the temperature of the reaction to 50°C and again checked for the formation of a precipitate. Formation of a precipitate indicated the substrate underwent nucleophilic substitution. The results observed for the Finkelstein reaction are shown in Table 1.

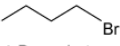
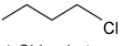
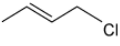
Table 1 Results of the S_N2 Reaction

Substrate	Result at 25°C	Result at 50°C
 1-Bromobutane	Precipitate	—
 1-Chlorobutane	No reaction	Moderate precipitate
 <i>tert</i> -Butyl chloride	No reaction	No reaction
 Crotyl chloride	Precipitate	—

Experiment 2

The students repeated the procedure followed in Experiment 1 except they added a few drops of each alkyl halide into a separate test tube containing 1 mL of 1% solution of AgNO₃ in ethanol at 25°C. The results observed for this reaction are shown in Table 2.

Table 2 Results of the S_N1 Reaction

Substrate	Result at 25°C	Result at 50°C
 1-Bromobutane	No reaction	Trace precipitate
 1-Chlorobutane	No reaction	No reaction
 <i>tert</i> -Butyl chloride	Heavy precipitate	—
 Crotyl chloride	Heavy precipitate	—

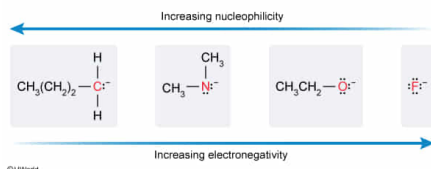
Suppose that NaI was replaced with another nucleophile in Experiment 1. Which of the following anions would be the most nucleophilic?

- ✓ ☒ A. $\text{CH}_3(\text{CH}_2)_2\text{CH}_2^-$ [22%]
- ☐ B. $(\text{CH}_3)_2\text{N}^-$ [10%]
- ☐ C. $\text{CH}_3\text{CH}_2\text{O}^-$ [22%]
- ☐ D. F^- [44%]

Correct

22% answered correctly

Explanation:



Nucleophiles donate electrons to **electrophiles** in nucleophilic substitutions. There are several factors that contribute to nucleophilicity, including charge and electronegativity. In general, as the **electronegativity** of a negatively charged atom **decreases** from **right to left** across a period on the periodic table, its **nucleophilicity increases** because less electronegative atoms stabilize a negative charge less effectively, have a weaker hold on electrons, and more readily donate electrons to an **electrophile**.

To compare the relative nucleophilicity of given anion, the electronegativity of the negatively charged atoms must be considered. The charged carbon atom of $\text{CH}_3(\text{CH}_2)_2\text{CH}_2^-$ is less electronegative than the nitrogen atom of $(\text{CH}_3)_2\text{N}^-$, the oxygen atom of $\text{CH}_3\text{CH}_2\text{O}^-$, and the fluorine atom of F^- . Therefore, electrons on a negatively charged carbon atom are less stabilized and more easily shared with an electrophile, making $\text{CH}_3(\text{CH}_2)_2\text{CH}_2^-$ the best nucleophile.

(Choices B, C, and D) Nucleophiles consisting of a negatively charged nitrogen, oxygen, or fluorine contain charged

atoms that are more electronegative and have a stronger hold on their electrons, making these species less nucleophilic than carbon nucleophiles.

Educational objective:

When comparing the nucleophilicity of atoms of equal negative charge, nucleophilicity tends to increase from right to left across a row of the periodic table as electronegativity decreases. Atoms of higher electronegativity more effectively stabilize negative charge and less readily donate electron density to electrophiles whereas those of lower electronegativity stabilize a negative charge less effectively and more readily donate electrons to electrophiles.

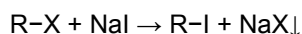
Time spent: 0 sec

Subject:
Organic Chemistry

QID: 401607

System:
Introduction to Organic Chemistry

A group of students studied the reactivity of four alkyl halides (R-X) in two nucleophilic substitution reactions. Nucleophilic substitution reactions can go through one of two mechanisms. The S_N2 reaction requires substitution of a leaving group with a nucleophile in a single step (Reaction 1), and the S_N1 reaction requires elimination of the leaving group and formation of an intermediate before nucleophilic substitution can occur, resulting in a two-step process (Reaction 2).



Reaction 1

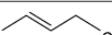


Reaction 2

Experiment 1

The students performed the S_N2 reaction (Reaction 1) by the addition of a few drops of each alkyl halide into a separate test tube containing 1 mL of 15% solution of NaI in acetone at 25°C. After mixing, they monitored the reaction for the formation of a precipitate (sodium halide salt). If no precipitate formed after 5 minutes, they raised the temperature of the reaction to 50°C and again checked for the formation of a precipitate. Formation of a precipitate indicated the substrate underwent nucleophilic substitution. The results observed for the Finkelstein reaction are shown in Table 1.

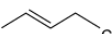
Table 1 Results of the S_N2 Reaction

Substrate	Result at 25°C	Result at 50°C
 1-Bromobutane	Precipitate	—
 1-Chlorobutane	No reaction	Moderate precipitate
 <i>tert</i> -Butyl chloride	No reaction	No reaction
 Crotyl chloride	Precipitate	—

Experiment 2

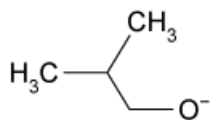
The students repeated the procedure followed in Experiment 1 except they added a few drops of each alkyl halide into a separate test tube containing 1 mL of 1% solution of AgNO₃ in ethanol at 25°C. The results observed for this reaction are shown in Table 2.

Table 2 Results of the S_N1 Reaction

Substrate	Result at 25°C	Result at 50°C
 1-Bromobutane	No reaction	Trace precipitate
 1-Chlorobutane	No reaction	No reaction
 <i>tert</i> -Butyl chloride	Heavy precipitate	—
 Crotyl chloride	Heavy precipitate	—

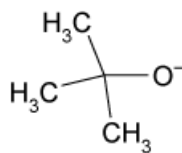
An alkoxide nucleophile is used instead of iodide in Experiment 1. Which of the alkoxides is the LEAST sterically hindered?

☐ A.



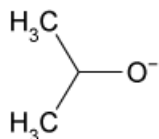
[4%]

☐ B.



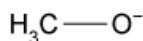
[1%]

☐ C.



[1%]

✓ ☒ D.

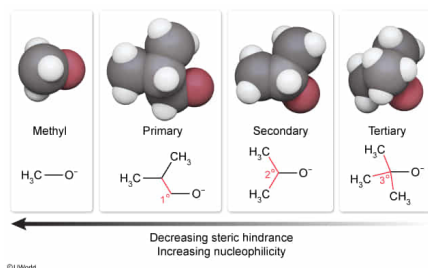


[92%]

Correct

92% answered correctly

Explanation:



The term **steric hindrance** denotes when a compound is bulky, which [keeps a reaction from occurring](#) or blocks the reaction from occurring at [specific locations](#) on a molecule. Steric hindrance plays a role in several organic reactions, including nucleophilic substitution and the formation of [kinetic and thermodynamic enolates](#). It is also a factor in reactivity comparisons of organic compounds, including [carboxylic acid derivatives](#), ketones and aldehydes, and nucleophiles.

To determine which of the given alkoxide nucleophiles is the least sterically hindered, the substitution of the carbon bonded to the negatively charged oxygen atom must be considered. More alkyl groups bonded to this carbon atom make it more substituted and, as a result, the nucleophilic site (the oxygen atom) becomes more sterically hindered (crowded). [Methoxide](#) has only three hydrogen atoms and no alkyl groups bonded to the carbon adjacent to the negatively charged oxygen. Because this carbon atom does not have any alkyl groups, it is the **least substituted** and therefore the **least sterically hindered** nucleophile.

(Choices A, B, and C) The carbon atoms bonded to the negatively charged oxygen in [isobutoxide](#), [isopropoxide](#), and [tert-butoxide](#) are a primary, secondary, and tertiary carbon, respectively. The increase in number of alkyl groups bonded to the carbon makes these nucleophiles more bulky and causes the nucleophilic atom in them to be more crowded, and therefore more sterically hindered than methoxide.

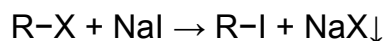
Educational objective:

Steric hindrance is a characteristic bulkiness in compound structures that either blocks a portion of a molecule so that it cannot react or keeps a reaction from occurring altogether. Decreasing the substitution on (or near) an atom decreases the steric hindrance of the atom.

Time spent:0 sec
Subject:
Organic Chemistry

QID:401608
System:
Introduction to Organic Chemistry

A group of students studied the reactivity of four alkyl halides (R-X) in two nucleophilic substitution reactions. Nucleophilic substitution reactions can go through one of two mechanisms. The S_N2 reaction requires substitution of a leaving group with a nucleophile in a single step (Reaction 1), and the S_N1 reaction requires elimination of the leaving group and formation of an intermediate before nucleophilic substitution can occur, resulting in a two-step process (Reaction 2).



Reaction 1

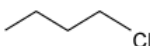
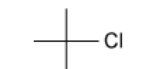
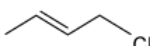


Reaction 2

Experiment 1

The students performed the S_N2 reaction (Reaction 1) by the addition of a few drops of each alkyl halide into a separate test tube containing 1 mL of 15% solution of NaI in acetone at 25°C. After mixing, they monitored the reaction for the formation of a precipitate (sodium halide salt). If no precipitate formed after 5 minutes, they raised the temperature of the reaction to 50°C and again checked for the formation of a precipitate. Formation of a precipitate indicated the substrate underwent nucleophilic substitution. The results observed for the Finkelstein reaction are shown in Table 1.

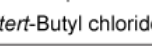
Table 1 Results of the S_N2 Reaction

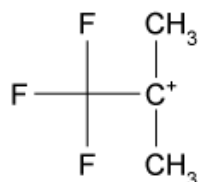
Substrate	Result at 25°C	Result at 50°C
 1-Bromobutane	Precipitate	—
 1-Chlorobutane	No reaction	Moderate precipitate
 tert-Butyl chloride	No reaction	No reaction
 Crotyl chloride	Precipitate	—

Experiment 2

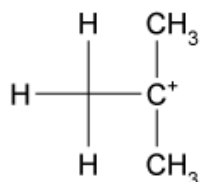
The students repeated the procedure followed in Experiment 1 except they added a few drops of each alkyl halide into a separate test tube containing 1 mL of 1% solution of AgNO₃ in ethanol at 25°C. The results observed for this reaction are shown in Table 2.

Table 2 Results of the S_N1 Reaction

Substrate	Result at 25°C	Result at 50°C
 1-Bromobutane	No reaction	Trace precipitate
 1-Chlorobutane	No reaction	No reaction
 <i>tert</i> -Butyl chloride	Heavy precipitate	—
 Crotyl chloride	Heavy precipitate	—



Compound 1



Compound 2

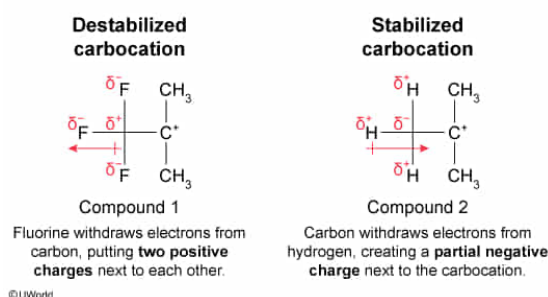
Compounds 1 and 2 are both intermediates that could form by a reaction with AgNO_3 in ethanol, but Compound 1 is less stable than Compound 2 because:

- ☐ A. Compound 1 contains an electron donating group that destabilizes the carbocation. [3%]
- ✓ ☒ B. there is an inductive effect from the fluorine atoms that destabilizes the carbocation. [89%]
- ☐ C. the methyl groups on the structure withdraw electrons from the positively charged carbon. [1%]
- ☐ D. the electronegative fluorine atoms cause the carbocation to rearrange into a secondary carbocation. [5%]

Correct

90% answered correctly

Explanation:



The **inductive effect** occurs with the donation of electron density through **sigma bonds**. **Electronegative** atoms or **electron withdrawing groups (EWG)** pull electrons away from an adjacent atom, creating a **dipole** with a partial negative charge on the electronegative atom and a partial positive charge on the adjacent atom. **Carbocations** are **stabilized** by **electron**

微信Mai_ange

donating groups (EDG), such as alkyl groups, because they donate electrons toward the positively charged carbon. **EWG** have the opposite effect: They tend to **destabilize** carbocations because the electronegative atom pulls electrons toward itself, creating a partial positive charge on the adjacent atom. With two adjacent positive charges, the carbocation is not stable.

Methyl ($-\text{CH}_3$) groups are EDG because the carbon atom is more electronegative than the adjacent hydrogen atom, creating a partial negative charge on carbon. This carbon can then donate electrons to the carbocation to stabilize it. The trifluoromethyl ($-\text{CF}_3$) group is an EWG because the fluorine atoms are more electronegative than the adjacent carbon and draw electrons toward themselves, creating a partial positive charge on the carbon next to the positively charged carbocation. Because Compound 1 has one EWG and Compound 2 does not contain any EWG, Compound 1 is less stable than Compound 2 due to the inductive effect from the fluorine atoms on $-\text{CF}_3$.

(Choices A and C) The $-\text{CH}_3$ groups in Compound 1 are EDG, which *donate* electrons to the carbocation and help *stabilize* the carbocation. It is the $-\text{CF}_3$ group that *withdraws* electrons and destabilizes the carbocation.

(Choice D) The fluorine atoms are highly electronegative and do not cause the carbocation to rearrange.

Educational objective:

The inductive effect occurs when electron density is donated through sigma bonds. Carbocations are stabilized by electron donating groups because they donate electrons to the positively charged carbon; they are destabilized by electron withdrawing groups because they pull electrons away from the carbocation, creating two adjacent positive charges.

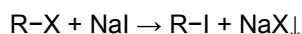
Time spent: 0 sec

Subject:
Organic Chemistry

QID: 401609

System:
Introduction to Organic Chemistry

A group of students studied the reactivity of four alkyl halides (R-X) in two nucleophilic substitution reactions. Nucleophilic substitution reactions can go through one of two mechanisms. The S_N2 reaction requires substitution of a leaving group with a nucleophile in a single step (Reaction 1), and the S_N1 reaction requires elimination of the leaving group and formation of an intermediate before nucleophilic substitution can occur, resulting in a two-step process (Reaction 2).



Reaction 1

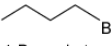
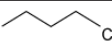
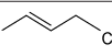


Reaction 2

Experiment 1

The students performed the S_N2 reaction (Reaction 1) by the addition of a few drops of each alkyl halide into a separate test tube containing 1 mL of 15% solution of NaI in acetone at 25°C. After mixing, they monitored the reaction for the formation of a precipitate (sodium halide salt). If no precipitate formed after 5 minutes, they raised the temperature of the reaction to 50°C and again checked for the formation of a precipitate. Formation of a precipitate indicated the substrate underwent nucleophilic substitution. The results observed for the Finkelstein reaction are shown in Table 1.

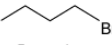
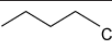
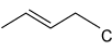
Table 1 Results of the S_N2 Reaction

Substrate	Result at 25°C	Result at 50°C
 1-Bromobutane	Precipitate	—
 1-Chlorobutane	No reaction	Moderate precipitate
 <i>tert</i> -Butyl chloride	No reaction	No reaction
 Crotyl chloride	Precipitate	—

Experiment 2

The students repeated the procedure followed in Experiment 1 except they added a few drops of each alkyl halide into a separate test tube containing 1 mL of 1% solution of AgNO₃ in ethanol at 25°C. The results observed for this reaction are shown in Table 2.

Table 2 Results of the S_N1 Reaction

Substrate	Result at 25°C	Result at 50°C
 1-Bromobutane	No reaction	Trace precipitate
 1-Chlorobutane	No reaction	No reaction
 <i>tert</i> -Butyl chloride	Heavy precipitate	—
 Crotyl chloride	Heavy precipitate	—

Which of the following statements is NOT supported by the results in Tables 1 and 2?

- ☐ A. In Experiment 2, crotyl chloride forms a stable carbocation. [8%]

- ☐ B. Primary alkyl halides react more readily in S_N2 reactions than in S_N1 reactions. [7%]
- ✓ ☒ C. Nucleophilic substitution reactions are not dependent on the substrate (alkyl halide) substitution. [70%]
- ☐ D. Bromide is a better leaving group than chloride and enhances reactivity in less favorable reactions. [12%]

Correct

70% answered correctly

Explanation:

S_N2 Conditions			S_N1 Conditions	
Reactions with NaI			Reactions with $AgNO_3$	
Substrate	Result at 25°C	Result at 50°C	Result at 25°C	Result at 50°C
1-Bromobutane	Precipitate	—	No reaction	Trace precipitate
1-Chlorobutane	No reaction	Moderate precipitate	No reaction	No reaction
<i>tert</i> -Butyl chloride	No reaction	No reaction	Heavy precipitate	—
Crotyl chloride	Precipitate	—	Heavy precipitate	—

R—Br reacts better than R—Cl

Primary and tertiary halides react differently

Primary alkyl halides react well in S_N2 conditions but not S_N1 conditions

Crotyl chloride reacts under S_N1 conditions

Results from experiments can be compared to **draw conclusions** about aspects of the reactions performed, including substrate reactivity, leaving group ability, and plausible intermediates formed. Observations for the S_N2 reaction conditions (Table 1) show that 1-bromobutane (a primary halide) formed a precipitate whereas *tert*-butyl chloride (a tertiary halide) did not react, even after heating. Conversely, results for the S_N1 reaction conditions (Table 2) record that *tert*-butyl chloride formed a heavy precipitate, but 1-bromobutane only formed a trace amount of precipitate after heating. This data demonstrates that S_N1 and S_N2 reactions *are* dependent on substrate substitution because primary and tertiary alkyl halides react only in certain conditions.

(Choice A) The passage states that Experiment 2 is an S_N1 reaction that involves carbocation formation. Because a heavy precipitate was observed for crotyl chloride, it must form a **resonance-stabilized carbocation** to react with $AgNO_3$.

This is an exception: Primary alkyl halides generally do not participate in S_N1 reactions, but crotyl chloride reacted because it is **allylic**.

(Choice B) 1-Bromobutane is a primary alkyl halide and formed a precipitate in the reaction with NaI (S_N2), but it formed only a minute amount in the reaction with $AgNO_3$ (S_N1), even after heating. Based on these observations, it can be concluded that primary alkyl halides undergo S_N2 reactions more readily than S_N1 reactions.

(Choice D) 1-Bromobutane and 1-chlorobutane are both primary alkyl halides that are structurally identical except for the halide leaving group. Based on the observations that 1-bromobutane formed a precipitate at room temperature but 1-chlorobutane had to be heated to form just a trace precipitate in an S_N2 reaction, it can be concluded that bromide is a better leaving group than chloride and that bromide comparatively enhances reactivity in less favorable reactions.

Educational objective:

Results from experiments can be compared to draw conclusions about aspects of the reactions performed, including the substrate reactivity, leaving group ability, and plausible intermediates formed. In particular, experimental results show that nucleophilic substitution reactions (S_N1 and S_N2) are dependent on substrate substitution.

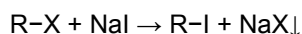
Time spent: 0 sec

Subject:
Organic Chemistry

QID: 401610

System:
Introduction to Organic Chemistry

A group of students studied the reactivity of four alkyl halides (R-X) in two nucleophilic substitution reactions. Nucleophilic substitution reactions can go through one of two mechanisms. The S_N2 reaction requires substitution of a leaving group with a nucleophile in a single step (Reaction 1), and the S_N1 reaction requires elimination of the leaving group and formation of an intermediate before nucleophilic substitution can occur, resulting in a two-step process (Reaction 2).



Reaction 1

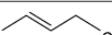


Reaction 2

Experiment 1

The students performed the S_N2 reaction (Reaction 1) by the addition of a few drops of each alkyl halide into a separate test tube containing 1 mL of 15% solution of NaI in acetone at 25°C. After mixing, they monitored the reaction for the formation of a precipitate (sodium halide salt). If no precipitate formed after 5 minutes, they raised the temperature of the reaction to 50°C and again checked for the formation of a precipitate. Formation of a precipitate indicated the substrate underwent nucleophilic substitution. The results observed for the Finkelstein reaction are shown in Table 1.

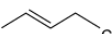
Table 1 Results of the S_N2 Reaction

Substrate	Result at 25°C	Result at 50°C
 1-Bromobutane	Precipitate	—
 1-Chlorobutane	No reaction	Moderate precipitate
 <i>tert</i> -Butyl chloride	No reaction	No reaction
 Crotyl chloride	Precipitate	—

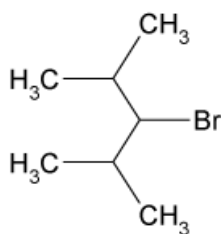
Experiment 2

The students repeated the procedure followed in Experiment 1 except they added a few drops of each alkyl halide into a separate test tube containing 1 mL of 1% solution of AgNO₃ in ethanol at 25°C. The results observed for this reaction are shown in Table 2.

Table 2 Results of the S_N1 Reaction

Substrate	Result at 25°C	Result at 50°C
 1-Bromobutane	No reaction	Trace precipitate
 1-Chlorobutane	No reaction	No reaction
 <i>tert</i> -Butyl chloride	Heavy precipitate	—
 Crotyl chloride	Heavy precipitate	—

If 3-bromo-2,4-dimethylpentane (shown below) was used as a substrate in the experiments described in the passage, what would the observation be for reaction with NaI at 25°C?

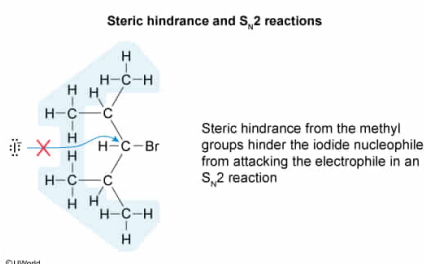


- ☐ A. Precipitate formation, because the substrate has a tertiary electrophile [5%]
☐ B. Precipitate formation, because the electrophilic carbon is not sterically hindered [6%]
☒ C. Trace precipitate, because the methyl groups hinder the addition of iodide to the electrophile [68%]
☐ D. No reaction, because S_N2 reactions cannot happen at sp^3 carbons [19%]

Correct

68% answered correctly

Explanation:



Steric hindrance arises when a molecule is large and bulky, and can **inhibit a reaction** from occurring or stop the reaction from taking place at a particular part of a molecule. In nucleophilic substitutions (S_N1 and S_N2), the nucleophile needs to be able to access the electrophilic carbon easily, but steric hindrance from the substrate or the nucleophile (or both) can prevent this.

The **extent of steric hindrance** a **substrate** has dictates in which **nucleophilic substitution** reaction it is likely to participate. Compounds with **tertiary leaving groups** (sterically hindered) are good substrates for S_N1 reactions because they involve formation of a **tertiary carbocation**, which is stabilized by three substituents. Molecules with primary leaving groups are generally good substrates for S_N2 reactions because the reaction is concerted and undergoes backside attack (nucleophile attacks on the opposite side of the leaving group). Molecules with secondary leaving groups are an intermediate case that can participate in S_N1 or S_N2 reactions, depending upon factors such as steric hindrance.

The passage states the reaction with NaI is an S_N2 reaction, indicating a concerted reaction occurs (ie, no carbocation formation or the possibility of a hydride shift). 3-Bromo-2,4-dimethylpentane is a secondary alkyl halide with two isopropyl groups bonded to the electrophile for the nucleophilic substitution reaction. Although secondary alkyl halides can undergo S_N2 reactions, the large bulky isopropyl groups will hinder the nucleophile's access to the electrophile. Therefore, 3-bromo-2,4-dimethylpentane will react very slowly with NaI at room temperature, forming only trace precipitate because the electrophile is too sterically hindered.

(Choices A and B) The central carbon of the isopropyl group (electrophile) is secondary rather than tertiary. Precipitate formation would be observed if the reaction occurred, but it does not because the electrophilic carbon is sterically hindered.

(Choice D) Although an S_N2 reaction does not occur between 3-bromo-2,4-dimethylpentane and NaI, S_N2 reactions can occur at **sp^3 carbons**.

Educational objective:

The steric hindrance of a substrate must be considered when determining whether it can undergo a particular reaction. The extent of a substrate's steric hindrance can dictate which nucleophilic substitution (S_N1 or S_N2) it is likely to undergo. S_N1 reactions permit sterically hindered electrophiles whereas S_N2 reactions require easily accessible electrophiles.

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